$$q_t := a_{0,t} - a_{1,t}$$

$$\hat{r}_i = \theta_t^T a_{i,t} - \beta_t^{(i)} \|a_{i,t}\|_{V_t^{-1}}$$

$$\hat{r}_1 \geq \hat{r}_0 \Rightarrow \theta_t^T q_t \leq \beta_t^{(1)} \|a_{1,t}\|_{V_t^{-1}} - \beta_t^{(0)} \|a_{0,t}\|_{V_t^{-1}}$$

$\because |\beta_t^{(i)}| \leq c_2 t^{\gamma_2}$

$$\therefore \theta_t^T q_t \leq c_2 t^{\gamma_2} \left(\|a_{0,t}\|_{V_t^{-1}} + \|a_{1,t}\|_{V_t^{-1}} \right) \quad (1)$$

$$\because \|\theta_* - \theta_t\|_{V_t} \leq c_1 t^{\gamma_1} g(\delta)$$

$$\therefore (\theta_* - \theta_t)^T q_t \leq c_1 t^{\gamma_1} g(\delta) \|q_t\|_{V_t^{-1}} \quad \text{(By Cauchy-Schwarz (2) in } V_t \text{ norm)}$$

$$(1),(2) \Rightarrow \theta_*^T q_t \leq c_2 t^{\gamma_2} \left(\|a_{0,t}\|_{V_t^{-1}} + \|a_{1,t}\|_{V_t^{-1}} \right) + c_1 t^{\gamma_1} g(\delta) \|a_{0,t} - a_{1,t}\|_{V_t^{-1}}$$

$\because \gamma_1, \gamma_2 \leq \gamma, \quad c_1, c_2 \leq c.$

$$\therefore \theta_*^T q_t \leq c t^{\gamma} \left[\left(\|a_{0,t}\|_{V_t^{-1}} + \|a_{1,t}\|_{V_t^{-1}} \right) + g(\delta) \|a_{0,t} - a_{1,t}\|_{V_t^{-1}} \right]$$

$$\leq c t^{\gamma} B_t$$

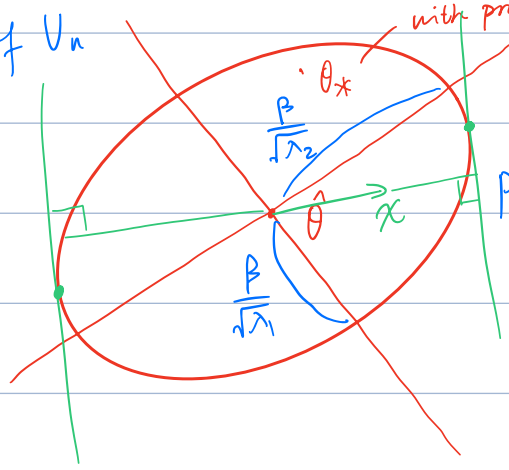
$\because \varepsilon_0$-dense $\quad \|\bar{q}_t - q_t\|_2 \leq 2\varepsilon_0.$

$$\theta_*^T \bar{q}_t \leq \theta_*^T q_t + 2\varepsilon_0 \leq B_t + 2\varepsilon_0$$

$\varepsilon \sim N(0, \sigma^2 I)$

$i \in [n]$

$\lambda_1 > \lambda_2$ of V_n



$$\beta = O\left(\sigma\left(\sqrt{d} + \sqrt{\log \frac{1}{\delta}}\right)\right)$$

$$C = \{\theta : (\theta - \hat{\theta})^T V_n (\theta - \hat{\theta}) \le \beta^2\} \qquad \theta_* \in C \text{ with prob } 1-\delta$$

$$C = \{\theta : \|\theta - \hat{\theta}\|_{V_n} \le \beta\}$$

pred: $\max_{\theta \in C} |x^T(\theta - \hat{\theta})| \le \|x\|_{V_n^{-1}} \|\theta - \hat{\theta}\|_{V_n}$

$$\le \|x\|_{V_n^{-1}} \beta$$

$$\|\theta_* - \hat{\theta}\|_2 \le \frac{O\left(\sigma\sqrt{d + \log \frac{1}{\delta}}\right)}{\sqrt{\lambda_{\min}(V_n)}}$$

$:= S_t$

If prior is $V_0 = \lambda I$, then $V_n = \lambda I + \sum x_i x_i^T$ $\quad \theta_n - \theta_* = V_n^{-1} \sum_i x_i \eta_i - \lambda V_n^{-1} \theta_*$

$$y_i = x_i^T \theta_* + \eta_i$$

$$\hat{\theta}_n = V_n^{-1} \sum_i x_i y_i = V_n^{-1}\left(\sum_i x_i x_i^T \theta_* + \sum_i x_i \eta_i\right)$$

$$V_n(\hat{\theta}_n - \theta_*) = V_n \hat{\theta}_n - V_n \theta_* = \sum_i x_i x_i^T \theta_* + \sum_i x_i \eta_i - V_n \theta_*$$

$$= (V_n - \lambda I)\theta_* + \sum_i x_i \eta_i - V_n \theta_*$$

$$= \sum_i x_i \eta_i - \lambda I \theta_*$$

$$\Rightarrow \hat{\theta}_n - \theta_* = V_n^{-1}\left(\sum_i x_i \eta_i - \lambda I \theta_*\right)$$

$\|a-b\| \le \|a\| + \|b\|$

$$\|\hat{\theta}_n - \theta_*\|_{V_n} \le \|V_n^{-1} \sum_i x_i \eta_i\|_{V_n} + \lambda \|V_n^{-1} \theta_*\|_{V_n}$$

$$\|V^{-1}z\|_V = \|z\|_{V^{-1}} \quad \Rightarrow \quad \leq \left\| \sum_i x_i \varepsilon_i \right\|_{V_n^{-1}} + \lambda \|\theta_*\|_{V_n^{-1}}$$

$$V_n \geq \lambda I \Rightarrow V_n^{-1} \preceq \frac{1}{\lambda} I \Rightarrow \|\theta_*\|_{V_n^{-1}}^2 = \theta_*^T V_n^{-1} \theta_* \leq \frac{1}{\lambda} \theta_*^T \theta_*$$

$$\|\hat{\theta}_N - \theta_*\|_2 \leq O\left(\frac{\sigma \sqrt{d + \log(1/\delta)}}{\sqrt{\lambda_{\min}(V_N)}} \right)$$

$$\Rightarrow \|\theta_*\|_{V_n^{-1}} \leq \frac{1}{\sqrt{\lambda}} \|\theta_*\|_2$$

$$\approx O\left(\sigma \sqrt{\frac{d + \log(1/\delta)}{N\kappa}} \right)$$

good and

$\kappa = \lambda_{\min}(\Sigma)$

$\Sigma = \mathbb{E}[x_i x_i^T]$, population covariance matrix

$$\Rightarrow \quad \leq \left\| \sum_i x_i \varepsilon_i \right\|_{V_n^{-1}} + \sqrt{\lambda} \|\theta_*\|_2$$

$$\forall \|\theta_*\|_2 \leq S \quad \Rightarrow \quad \leq \underbrace{\left\| \sum_i x_i \varepsilon_i \right\|_{V_n^{-1}}}_{\text{statistical noise}} + \underbrace{\sqrt{\lambda} S}_{\text{regularization bias}}$$

Online Learning $t \in [t]$

Standard self-normalized bound / inequality

$\varepsilon_i \sim N(0, R^2)$ $(R = \sigma)$

$$\left\| \sum_i x_i \varepsilon_i \right\|_{V_t^{-1}} \leq R \sqrt{2 \log \left[\frac{\det(V_t)^{\frac{1}{2}}}{\det(\lambda I)^{\frac{1}{2}} \delta} \right]}$$

V_0

$$\|\hat{\theta}_t - \theta_*\|_{V_t} \leq \beta_t \quad \Rightarrow \quad \beta_t = R \sqrt{2 \log \left[\frac{\det(V_t)^{\frac{1}{2}}}{\lambda^{\frac{d}{2}} \delta} \right]} + \sqrt{\lambda} S$$

Assume $\|x_t\|_2 \leq L$. $\quad \operatorname{tr}(V_t) = \lambda d + \sum_i \|x_i\|^2 \leq \lambda d + t L^2$

$$\det(V_t) = \prod_{j=1}^{d} \mu_j \leq \left(\frac{\sum_j \mu_j}{d} \right)^d = \left(\frac{\operatorname{tr}(V_t)}{d} \right)^d \leq \left(\lambda + \frac{tL^2}{d} \right)^d$$

Aside

$\operatorname{tr}(V_t) = \operatorname{tr}(Q \Lambda Q^T) = \operatorname{tr}(\Lambda Q^T Q)$

$= \operatorname{tr}(\Lambda I) = \operatorname{tr}(\Lambda) = \sum_j \mu_j$

$$\beta_t \leq R \sqrt{2 \log \left[\frac{\left(\lambda + \frac{tL^2}{d}\right)^{\frac{d}{2}}}{\lambda^{\frac{d}{2}} \delta} \right]} + \sqrt{\lambda} S$$

$$= R \sqrt{2 \log \left[\left(1 + \frac{tL^2}{\lambda d}\right)^{\frac{d}{2}} / \delta \right]} + \sqrt{\lambda} S$$

$$= R \sqrt{d \log\left(1 + \frac{tL^2}{\lambda d}\right) + 2 \log \frac{1}{\delta}} + \sqrt{\lambda} S$$

$$\leq R\sqrt{d\log\left(1+\frac{tL^2}{\lambda d}\right)+\log\frac{1}{\delta}}+\sqrt{\lambda}S$$

$$\beta_t = O\left(R\sqrt{d\log t+\log\frac{t}{\delta}}\right)$$

Derivation of self-normalized martingale inequality

$S_t := \sum_{i=1}^{t} x_i\eta_i$ $\quad \hat{\theta}_t - \theta_* = V_t^{-1}S_t - V_t^{-1}V_0\theta_*$

Interlude: sub-Gaussian noise

MGF moment generating function

$Z\sim N(0,R^2) \Rightarrow \mathbb{E}[e^{\lambda Z}] = e^{\lambda^2R^2/2}$

Sub-Gaussian: $\mathbb{E}[e^{\lambda\eta}] \leq \mathbb{E}[e^{\lambda Z}],\ \forall\lambda\in\mathbb{R}$

control the tail (where η is large)

Core assumption: $\mathcal{F}_0\subseteq\mathcal{F}_1\subseteq\cdots\subseteq\mathcal{F}_t$ $\quad x_t\in\mathcal{F}_{t-1}$ (x_t is dependent on the past info)

$\mathbb{E}[e^{\lambda\eta_t}\mid\mathcal{F}_{t-1}] \leq e^{\lambda^2R^2/2},\ \forall\lambda\in\mathbb{R}$

$\mathcal{F}_{t-1}$: all info before η_t appears in epoch t.

arbitrary $q\in\mathbb{R}^d$, $q^TS_t = \sum_{i=1}^{t}(q^Tx_i)\eta_i$ $\quad x_t\in\mathcal{F}_{t-1}$, $\eta_t\notin\mathcal{F}_{t-1}$

$\eta_t\sim N(0,R^2)$

$\mathbb{E}[e^{(q^Tx_t)\eta_t}\mid\mathcal{F}_{t-1}] \leq \exp\left[\frac{R^2}{2}(q^Tx_t)^2\right]$

writ. dist of η_t

$\mathbb{E}\left[\exp\left[(q^Tx_t)\eta_t - \frac{R^2}{2}(q^Tx_t)^2\right]\,\middle|\,\mathcal{F}_{t-1}\right] \leq 1.$

$A_t := \sum_{i=1}^{t} x_ix_i^T$ $\quad q^TA_tq = \sum_{i=1}^{t}(q^Tx_i)^2$

Definition $M_t(q) = \exp\left(q^TS_t - \frac{R^2}{2}q^TA_tq\right)$

$\frac{M_t(q)}{M_{t-1}(q)} = \exp\left(q^T(S_t-S_{t-1}) - \frac{R^2}{2}(q^TA_tq - q^TA_{t-1}q)\right)$

$S_t = S_{t-1} + X_t \varepsilon_t$

$A_t = A_{t-1} + X_t X_t^T$

$= \exp\left((q^T X_t)\varepsilon_t - \frac{R^2}{2}(q^T X_t)^2 \right)$

$$\mathbb{E}\left[\frac{M_t(q)}{M_{t-1}(q)} \mid \mathcal{F}_{t-1} \right] \le 1$$

Since $M_{t-1}(q)$ is known given $\mathcal{F}_{t-1}$,

$$\mathbb{E}[M_t(q) \mid \mathcal{F}_{t-1}] \le M_{t-1}(q)$$

Let $q \sim N\left(0, \frac{1}{R^2} V_0^{-1}\right)$

$$p(q) = \frac{\det(R^2 V_0)^{\frac{1}{2}}}{(2\pi)^{d/2}} \exp\left(-\frac{R^2}{2} q^T V_0 q \right)$$

$$\bar{M}_t := \int M_t(q)\, p(q)\, dq$$

$$= \sqrt{\frac{\det V_0}{\det V_t}} \exp\left(\frac{\|S_t\|^2_{V_t^{-1}}}{2R^2} \right)$$

$\mathbb{E}[\bar{M}_t \mid \mathcal{F}_{t-1}] \le \bar{M}_{t-1}$

bc $p(q)$ are the same.

Ville inequality

$M_0(q) = 1$, $\forall q$ bc

$S_0 = 0$ $A_0 = 0$

$\Rightarrow \bar{M}_0 = 1$

$\Rightarrow \mathbb{E}[\bar{M}_0] = 1$

$$\tau := \inf\left\{ t : M_t \ge \frac{1}{\delta} \right\}, \quad r \wedge n = \min\{r, n\}$$

$$\frac{1}{\delta} P(\tau \le n) = \frac{1}{\delta} \mathbb{1}[\tau \le n] \le \mathbb{E}[M_{\tau \wedge n}] \le \mathbb{E}[M_0] \le 1 \Rightarrow P(\tau \le n) \le \delta$$

$$P\left(\sup_{t \ge 0} M_t \ge \frac{1}{\delta} \right) \le \delta \Rightarrow \text{For prob} = 1-\delta, \ \bar{M}_t < \frac{1}{\delta}, \forall t$$

$$\bar{M}_t < \frac{1}{\delta}, \forall t \Rightarrow \|S_t\|_{V_t^{-1}} \le R \sqrt{2 \log\left[\frac{\det(V_t)^{\frac{1}{2}}}{\det(V_0)^{\frac{1}{2}} \delta} \right]} \text{ with prob } 1-\delta, \forall t$$

10. Now Let's Actually Evaluate the Gaussian Integral

Substituting the Gaussian prior into the mixture, we obtain

$$\bar{M}_t = \int \exp\left(q^\top S_t - \frac{R^2}{2} q^\top A_t q \right) p(q)\, dq.$$

Recall that

$$A_t + V_0 = V_t.$$

Therefore, combining the exponential terms gives

$$q^\top S_t - \frac{R^2}{2} q^\top V_t q.$$

Hence,

$$\bar{M}_t = C \int \exp\left[q^\top S_t - \frac{R^2}{2} q^\top V_t q \right] dq,$$

where the normalization constant is

$$C = \frac{\det(R^2 V_0)^{1/2}}{(2\pi)^{d/2}}.$$

11. Completing the Square

Consider the exponent

$$q^\top S_t - \frac{R^2}{2} q^\top V_t q.$$

Define

$$m_t = \frac{1}{R^2} V_t^{-1} S_t.$$

We can then complete the square:

$$q^\top S_t - \frac{R^2}{2} q^\top V_t q$$
$$= -\frac{R^2}{2}(q - m_t)^\top V_t (q - m_t) + \frac{1}{2R^2} S_t^\top V_t^{-1} S_t.$$

Substituting this expression back into the integral yields

$$\bar{M}_t = C \exp\left(\frac{1}{2R^2} S_t^\top V_t^{-1} S_t \right) \times \int \exp\left[-\frac{R^2}{2}(q - m_t)^\top V_t (q - m_t) \right] dq.$$

Now we use the standard multivariate Gaussian integral identity:

$$\int_{\mathbb{R}^d} \exp\left[-\frac{R^2}{2}(q-m)^\top V_t (q-m) \right] dq = \frac{(2\pi)^{d/2}}{\det(R^2 V_t)^{1/2}}.$$

This holds because V_t is positive definite.

Substituting this result, we get

$$\bar{M}_t = \frac{\det(R^2 V_0)^{1/2}}{(2\pi)^{d/2}} \exp\left(\frac{1}{2R^2} S_t^\top V_t^{-1} S_t \right) \times \frac{(2\pi)^{d/2}}{\det(R^2 V_t)^{1/2}}.$$

The factors of $(2\pi)^{d/2}$ cancel. Moreover, since

$$\det(R^2 V) = R^{2d} \det(V),$$

the factors involving R^{2d} also cancel.

We therefore arrive at

$$\boxed{\bar{M}_t = \sqrt{\frac{\det V_0}{\det V_t}} \exp\left(\frac{\|S_t\|^2_{V_t^{-1}}}{2R^2} \right)}$$

where

$$\|S_t\|^2_{V_t^{-1}} = S_t^\top V_t^{-1} S_t.$$

This is the most elegant step in the entire proof of the self-normalized concentration theorem.

Elliptical Potential Lemma

$$\det(A + uv^T) = \det(A)\det(I + A^{-1}uv^T)$$

$$= \det(A)(1 + v^T A^{-1} u)$$

$$\det(V_t) = \det(V_{t-1} + x_t x_t^T) = \det(V_{t-1}) \cdot \left(1 + \|x_t\|^2_{V_{t-1}^{-1}}\right)$$

(def: $u_t = \|x_t\|^2_{V_{t-1}^{-1}}$)

$$= \det(V_{t-1})(1 + u_t)$$

$$\prod_{t=1}^{T} 1 + u_t = \frac{\det V_T}{\det V_0} \Rightarrow \sum_{t=1}^{T} \log(1 + u_t) = \log \frac{\det V_T}{\det V_0}$$

$$\sum_{t=1}^{T} \min\{1, u_t\} \le \sum_{t=1}^{T} 2\log(1 + u_t) = 2\log \frac{\det V_T}{\det V_0}$$

$$\|\hat{\theta}_{t-1} - \theta_*\|_{V_{t-1}} \le \beta_{t-1}$$

$$|x^T(\hat{\theta}_{t-1} - \theta_*)| \le \|x\|_{V_{t-1}^{-1}} \|\hat{\theta}_{t-1} - \theta_*\|_{V_{t-1}} \le \beta_{t-1} \|x_t\|_{V_{t-1}^{-1}}$$

$$x_t^* := \arg\max_x x^T \theta_*$$

$x_t = \max_x$ UCB-score

$$(x_t^*)^T \theta_* \le (x_t^*)^T \hat{\theta}_{t-1} + \beta_{t-1} \|x_t^*\|_{V_{t-1}^{-1}} \le x_t^T \hat{\theta}_{t-1} + \beta_{t-1} \|x_t\|_{V_{t-1}^{-1}}$$

$$x_t^T \theta_* \ge x_t^T \hat{\theta}_{t-1} - \beta_{t-1} \|x_t\|_{V_{t-1}^{-1}}$$

$$\Rightarrow (x_t^*)^T \theta_* - x_t^T \theta_* \le 2\beta_{t-1} \|x_t\|_{V_{t-1}^{-1}}$$

$$r_t \le 2\beta_{t-1} \|x_t\|_{V_{t-1}^{-1}}$$

$$R_T = \sum_{t=1}^{T} r_t \le \sum_{t=1}^{T} 2\beta_{t-1} \|x_t\|_{V_{t-1}^{-1}} \le 2\beta_T \sum_{t=1}^{T} \|x_t\|_{V_{t-1}^{-1}}$$

Cauchy-Schwarz

$$\leq 2\beta_T \sqrt{T \sum_{t=1}^{T} \|x_t\|^2_{V_{t-1}^{-1}}} \leq 2\beta_T \sqrt{T \cdot 2\log \frac{\det V_T}{\det V_0}}$$

$$= 2\left(R\sqrt{2\log\left[\frac{\det(V_T)^{\frac{1}{2}}}{\det(V_0)^{\frac{1}{2}}\delta}\right]} + \sqrt{\lambda}S\right)\sqrt{T \cdot 2\log \frac{\det V_T}{\det V_0}}$$

$\log \frac{\det V_T}{\det V_0} \leq d \log T$

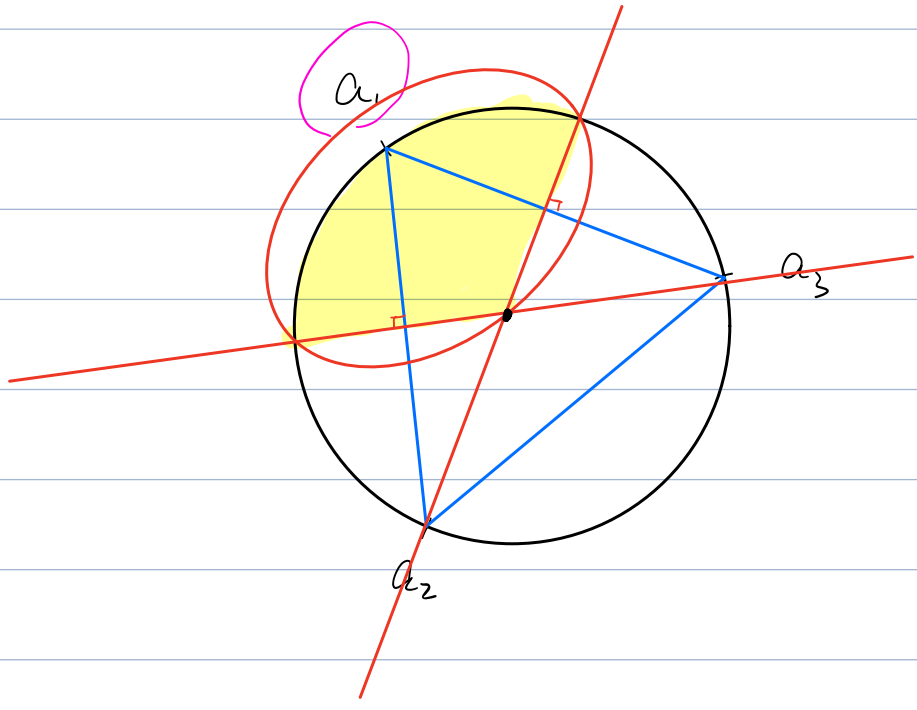
$$\leq 2\left(R\sqrt{d\log T + \log\tfrac{1}{\delta}} + \sqrt{\lambda}S\right)\sqrt{2dT\log T}$$

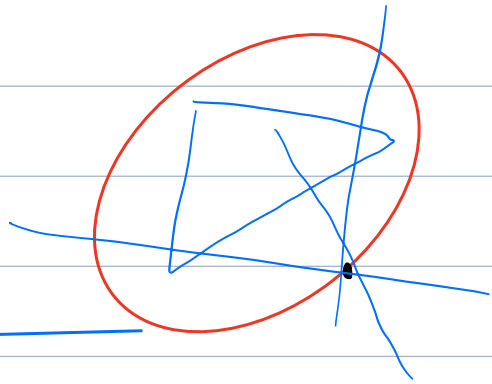
$$= O\left(\sqrt{d\log T + \log\tfrac{1}{\delta}}\ \sqrt{dT\log T}\right)$$

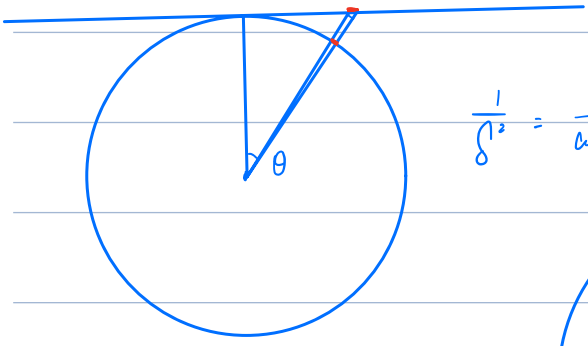
$$\approx \tilde{O}\left(d\sqrt{T}\right)$$

$$\|\hat{\theta}_T - \theta_*\|_2 \leq O\left(\sigma\sqrt{\frac{d\log T + \log(1/\delta)}{\lambda_{\min}(V_T)}}\right)$$

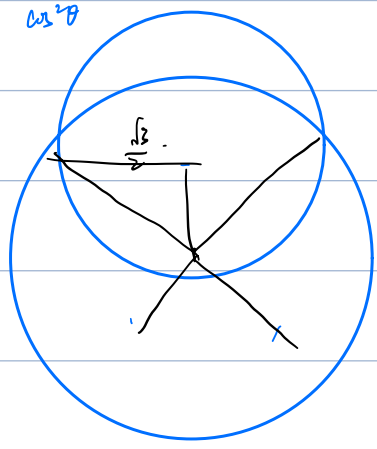
$$\overset{\text{good cond}}{\approx} O\left(\sigma\sqrt{\frac{d\log t + \log(1/\delta)}{Tk}}\right)$$

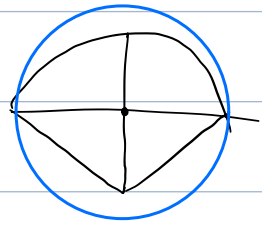


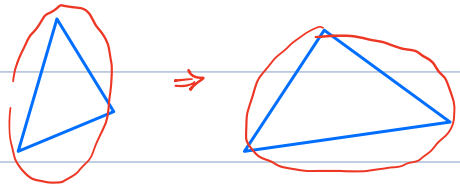


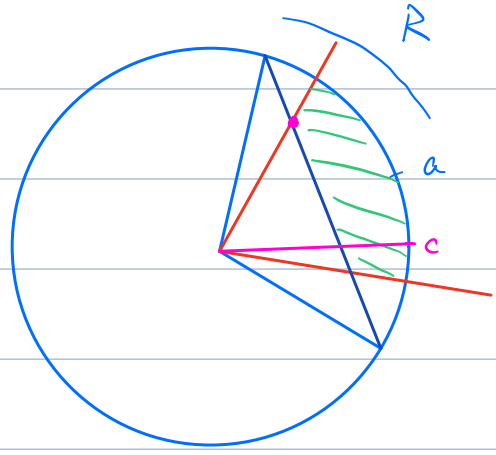


$$\frac{1}{\delta^2} = \frac{1}{\cos^2\theta}$$









$$\left(cK^{-\frac{1}{d-1}}\right)^T \geq \varepsilon$$

$$T\left(\log c - \frac{1}{d-1}\log K\right) \geq \log \varepsilon$$

$$T \geq \frac{\log \varepsilon}{\log c - \frac{\log K}{d-1}}$$

$$\left(cK^{-\frac{1}{d-1}}\right)^T = \frac{1}{K} \Rightarrow T = \frac{\log K}{\frac{1}{d-1}\log K - \log c}$$

$$\frac{1}{d-1} - \frac{\log c}{\log K} \Leftrightarrow \frac{1}{d}$$

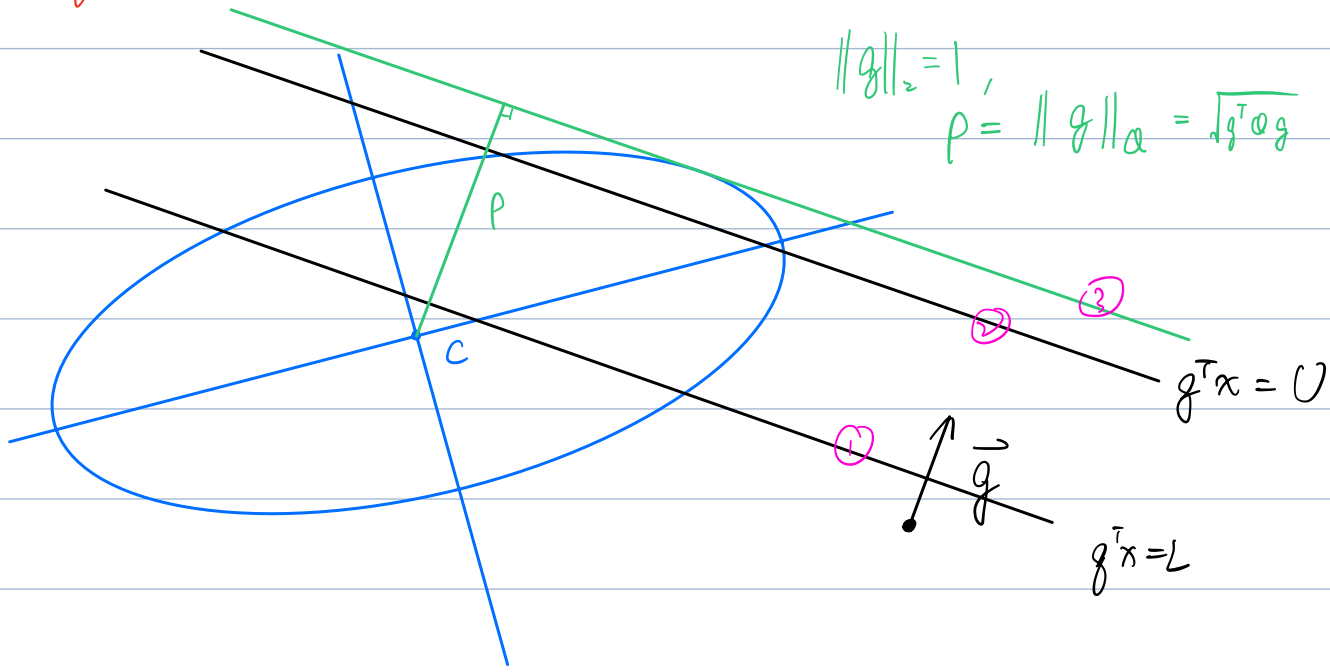
$$\updownarrow$$

$$d$$

$$\frac{m}{q^2}\left(\frac{\alpha}{\tan\alpha}\right)^2\left(1+O(q^{-2})\right) < \frac{m}{q^2}\left(1+\Theta(q^{-2})\right)$$

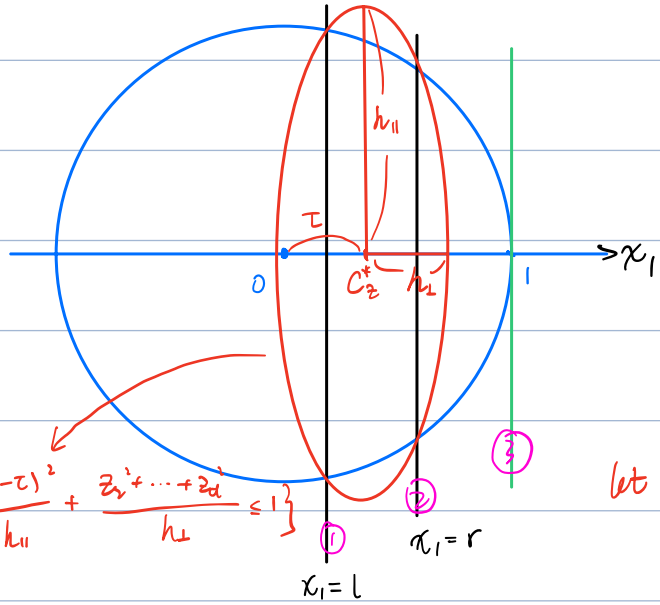
$$\approx \frac{m}{q^2}$$

$$E=\{x:(x-c)^T Q^{-1}(x-c)\le 1\}$$



↓ Whitening

$$l=\frac{L-g^Tc}{\rho},\quad r=\frac{U-g^Tc}{\rho}$$



$$\mathrm{Vol}(E_z^*)=\kappa_d\sqrt{h_{11}h_\perp^{d-1}}$$

$$\frac{(l-\tau)^2}{h_{11}}+\frac{1-l^2}{h_\perp}=1$$

$$\frac{(r-\tau)^2}{h_{11}}+\frac{1-r^2}{h_\perp}=1$$

$$E_z^*=\left\{z:\frac{(z_1-\tau)^2}{h_{11}}+\frac{z_2^2+\cdots+z_d^2}{h_\perp}\le 1\right\}$$

let $p=lr$, $s=l+r$

$$\Rightarrow \begin{cases} h_\perp = h_{11} \cdot \dfrac{s}{s-2\tau} \\ h_{11} = 1+\tau^2 - \dfrac{2\tau(1+\rho)}{s} \end{cases}$$

$$\text{Vol} \propto h_{11}\, h_\perp^{d-1}$$

$$= h_{11}^{d} \cdot \left(\frac{s}{s-2\tau}\right)^{d-1}$$

$$\log \text{Vol} = d \log h_{11} - (d-1) \log |s-2\tau| + \text{const.}$$

$$\begin{cases} \tau = \dfrac{d(1+r)^2 + 2(1+lr) - \sqrt{\Delta}}{2(d+1)(l+r)}, \quad \Delta = d^2(r^2-l^2)^2 + 4(1-l^2)(1-r^2) \\ h_{11} = d(\tau - l)(\tau - r) \\ h_\perp = \dfrac{1-l^2}{1 - \frac{(l-\tau)^2}{h_{11}}} = \dfrac{1-r^2}{1-\frac{(r-\tau)^2}{h_{11}}} \end{cases}$$

If $l = -r = b$, $h_{11} = db^2$, $h_\perp = \dfrac{d(1-b^2)}{d-1}$, $\tau = 0$

Undo the whitening

$$Q = LL^T$$

Whitening: $\rightsquigarrow z = L^{-1}(x-c) \Rightarrow x = c + Lz$

Since $(x-c)^T Q^{-1} (x-c) = (x-c)^T L^{-T} L^{-1} (x-c) = z^T z \le 1$.

$$E_z^* = \{ z : (z-a)^T D^{-1} (z-a) \le 1 \}$$

$$z - a = L^{-1}(x - c_+) \Rightarrow c_+ = c + La$$

$$\left[L^{-1}(x-c_+)\right]^T D^{-1} \left[L^{-1}(x-c_+)\right] \le 1$$

$$(x-c_+)^T L^{-T} D^{-1} L^{-1} (x-c_+) \le 1$$

$$L^{-T} D^{-1} L^{-1} = (LDL^T)^{-1} = Q_+^{-1} \Rightarrow Q_+ = LDL^T$$

We don't want to rotate, actually (I)

$$D = \begin{pmatrix} h_{11} & & \\ & h_\perp & \\ & & \ddots \\ & & & h_\perp \end{pmatrix} = h_{11} e_1 e_1^T + h_\perp (I - e_1 e_1^T)$$

$$= h_\perp I + (h_{11} - h_\perp) e_1 e_1^T \Rightarrow h_\perp I + (h_{11} - h_\perp) n n^T$$

$$g^T x = U \Rightarrow g^T (c + Lz) = U$$

$$g^T c + (L^T g)^T z = U$$

$$(L^T g)^T z = U - g^T c$$

Suppose $g \xrightarrow{L^T} \alpha n$ n is a unit vector

$$g^T (x - c) = (\alpha n)^T z, \quad \Rightarrow \alpha n = L^T g,$$

$$n = \frac{L^T g}{\|L^T g\|} = \frac{L^T g}{\sqrt{g^T L L^T g}} = \frac{L^T g}{\sqrt{g^T Q g}} = \frac{L^T g}{\|g\|_Q}$$

$$c_+ = c + La = c + L \tau n = c + \tau L \frac{L^T g}{\|L^T g\|} = c + \tau \frac{Qg}{\sqrt{g^T Q g}}$$

$$Q_+ = L D L^T = L h_\perp I L^T + L (h_{11} - h_\perp) n n^T L^T$$

$$= h_\perp Q + (h_{11} - h_\perp) \cdot Ln (Ln)^T = h_\perp Q + (h_{11} - h_\perp) \frac{Qg}{\sqrt{g^T Q g}} \left(\frac{Qg}{\sqrt{g^T Q g}}\right)^T$$

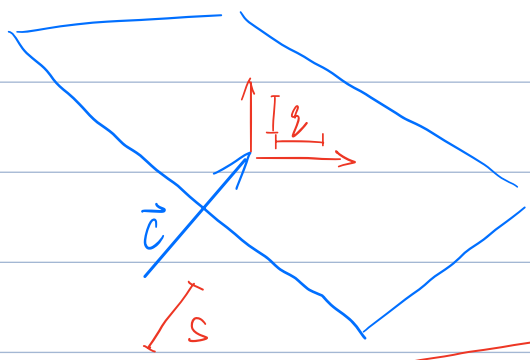
$$= h_\perp Q + (h_{11} - h_\perp) \frac{Q g g^T Q}{g^T Q g}$$

$$c_+ = c + \tau \frac{Qg}{\rho}$$

$$Q_+ = h_\perp Q + (h_{11} - h_\perp) \frac{Q g g^T Q}{\rho^2} \qquad (Q^T = Q)$$

$$\rho = \sqrt{g^T Q g}$$

$D-1=d$



$x \in \mathbb{R}^D$.

write x as $x = sc + Fq$

$s = c^T x$, $q = F^T x$ $\quad q \in \mathbb{R}^d$

$y^T W^{-1} y \le 1 \iff q^T W^{-1} q \le s^2$

$E(a,H) = \{y : (y-a)^T H^{-1} (y-a) \le 1\}$ $\quad y \in \mathbb{R}^d$.

The ray through $c+Fy$ is $x = s(c+Fy)$, $s \ge 0$

Let $q = sy$. $\quad x = sc + Fq$

$$\left(\frac{q}{s} - a\right)^T H^{-1} \left(\frac{q}{s} - a\right) \le 1$$

$$(q - sa)^T H^{-1} (q - sa) \le s^2.$$

The cone generated by $E(a,H)$ (tangent ellipsoid)

is

$$K(a,H) = \{sc + Fq : (q-sa)^T H^{-1} (q-sa) \le s^2, \ s \ge 0\}$$

$$\Downarrow$$

$$\left(F^T x - (c^T x)\, a\right)^T H^{-1} \left(F^T x - (c^T x)\, a\right) \le (c^T x)^2$$

Let $w = \begin{pmatrix} s \\ q \end{pmatrix} \in \mathbb{R}^D$

↳ is equivalent to $w^T M^{-1} w \geq 0$. what's M?

$$L = \begin{pmatrix} 1 & 0 \\ a & Id \end{pmatrix} \qquad L^{-1} = \begin{pmatrix} 1 & 0 \\ -a & Id \end{pmatrix}$$

$$M = L \begin{pmatrix} 1 & 0 \\ 0 & -H \end{pmatrix} L^T \Rightarrow M^{-1} = L^{-T} \begin{pmatrix} 1 & 0 \\ 0 & -H^{-1} \end{pmatrix} L^{-1}$$

$$L^{-1} \begin{pmatrix} s \\ q \end{pmatrix} = \begin{pmatrix} s \\ q - sa \end{pmatrix}, \quad w^T M^{-1} w = \begin{pmatrix} s \\ q-sa \end{pmatrix}^T \begin{pmatrix} 1 & 0 \\ 0 & -H^{-1} \end{pmatrix} \begin{pmatrix} s \\ q-sa \end{pmatrix}$$

$$\begin{pmatrix} 1 & a^T \\ a & aa^T - H \end{pmatrix} \qquad = s^2 - (q-sa)^T H^{-1} (q-sa)$$

$$M = U \begin{pmatrix} \mu & 0 \\ 0 & -\mathrm{diag}(\nu_1, \dots, \nu_d) \end{pmatrix} U^T$$

Sylvester's law of inertia

since $H \succ 0$

$-H \prec 0$

$$M^{-1} = U \begin{pmatrix} 1/\mu & 0 \\ 0 & -\mathrm{diag}(1/\nu_1, \dots, 1/\nu_d) \end{pmatrix} U^T$$

$w^T M^{-1} w \geq 0 \qquad w^T U \begin{pmatrix} \quad \\ \quad \end{pmatrix} U^T w \geq 0 \qquad w' = U^T w$

$$w' = \begin{pmatrix} w_0' \\ w_1' \\ \vdots \\ w_d' \end{pmatrix}$$

$$\Rightarrow \frac{(w_0')^2}{\mu} - \sum_{j=1}^{d} \frac{(w_j')^2}{\nu_j} \geq 0.$$

$$\sum_{j=1}^{d} \frac{(w_j')^2}{\nu_j/\mu} \leq (w_0')^2$$

$$w' = \begin{pmatrix} s_+ \\ q_+ \end{pmatrix}$$

$q_+^T W_+^{-1} q_+ \leq s_+^2$

old frame	new frame.
$w = (s, q)$	$w' = (s_+, q_+)$
$x = (c, F)\, w$	$x = (c_+, F_+)\, w'$
$q^T W^{-1} q \leq s^2$	$q_+^T W_+^{-1} q_+ \leq s_+^2$

$$W_+ = \operatorname{diag}\left(\frac{\nu_1}{\mu}, \cdots, \frac{\nu_d}{\mu}\right)$$

$$\chi = [c \; F]\begin{pmatrix} s \\ v \end{pmatrix} = [c \; F]\, W$$

$$U = [u_0 \;\; U_-]$$

$$c_+ = [c \; F]\, u_0 \qquad F_+ = [c \; F]\, U_-$$

$$[C_+ \; F_+] = [c \; F]\, U$$

$[C_+ F_+] W' = [c F] U U^T W = [c F] W$, $[c \; F]$, U are orthonormal

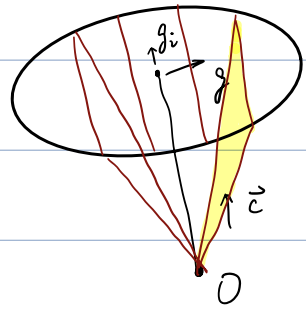
So $\|c_+\| = 1$ $\quad (F_+)^T (F_+) = I$. $\quad F_+^T C_+ = 0$

$\Rightarrow$ c_+, F_+ are exactly the new ellipsoidal-cone parameters.

$$\begin{cases} c_+ = [c \; F]\, u_0 \\ F_+ = [c \; F]\, U_- \\ W_+ = \operatorname{diag}\left(\frac{\nu_1}{\mu}, \cdots, \frac{\nu_d}{\mu}\right) \end{cases}$$

$$Qg = \lambda g \qquad \|g\|_2 = 1$$

$$n = \frac{L^T g}{\sqrt{g^T Q g}} = \frac{\sqrt{\lambda}\, g}{\sqrt{\lambda}} = g$$



$$a = \tau n = \tau\sqrt{\lambda}\, g$$

$$H = Q_t = h_\perp Q + (h_{11} - h_\perp)\frac{Qg\,g^T Q}{g^T Q g}$$

$$= h_\perp Q + (h_{11} - h_\perp)\,\lambda g g^T$$

$$\Rightarrow Hg = \lambda h_{11}\, g$$

For other eigenvectors g_i of Q.

$$Qg_i = \lambda_i g_i \,, \quad g_i^T g = 0$$

$$\Rightarrow Hg_i = \lambda_i h_\perp g_i$$

$g \in \mathbb{R}^d$

$\bar{g} \in \mathbb{R}^D$

$$\bar{g} = Fg \,, \quad \bar{g}_i = F g_i$$

Use the coordinate order $(c, \bar{g}, \bar{g}_2, \ldots, \bar{g}_d)$

$$F = [\bar{g}\ \bar{g}_2\ \cdots\ \bar{g}_d]$$

↑ Let coordinates align with eigenvectors of ellipsoid

$$a = \begin{pmatrix} \tau\sqrt{\lambda} \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^d$$

$$M = \begin{pmatrix} 1 & a^T \\ a & aa^T - H \end{pmatrix} = \begin{pmatrix} 1 & \tau\sqrt{\lambda} & & & \\ \tau\sqrt{\lambda} & (\tau\sqrt{\lambda})^2 - \lambda h_{11} & & & \\ & & -\lambda_2 h_\perp & & \\ & & & \ddots & \\ & & & & -\lambda_d h_\perp \end{pmatrix}$$

$\tau\sqrt{\lambda} = \xi \qquad \lambda h_{11} = p$

$$M_2 = \begin{pmatrix} 1 & \xi \\ \xi & \xi^2 - p \end{pmatrix} = U_2 \begin{pmatrix} \mu & 0 \\ 0 & -\nu \end{pmatrix} U_2^T$$

$$U_2 = \frac{1}{\sqrt{1+\zeta^2}} \begin{pmatrix} 1 & -\zeta \\ \zeta & 1 \end{pmatrix}$$

$$\mu = \frac{\chi + \sqrt{\chi^2 + 4p}}{2} \qquad \chi = 1 + \xi^2 - p$$

$$-\nu = \frac{\chi - \sqrt{\chi^2 + 4p}}{2}$$

$$\mu\nu = p \qquad \mu + \nu = \chi$$

$$\zeta = \frac{\mu - 1}{\xi}$$

$$U = \begin{pmatrix} U_2 & \\ & I_{D-2} \end{pmatrix}$$

$$M = U \; \mathrm{diag}(\mu_1, -\nu, -\lambda_1 h_1, \ldots, -\lambda_d h_1) \; U^T$$

$$C_+ = [C \; F] \, U_0 = [c \; \bar{g} \; \bar{g}_2 \cdots \bar{g}_d] \, \frac{1}{\sqrt{1+\zeta^2}} \begin{bmatrix} 1 \\ \zeta \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

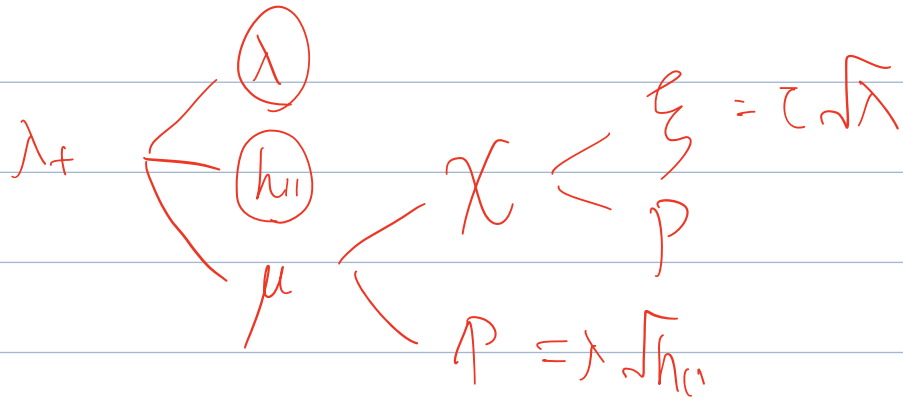
$$= \frac{c + \zeta \bar{g}}{\sqrt{1+\zeta^2}}$$

$$\bar{g}_+ = \frac{-\zeta c + \bar{g}}{\sqrt{1+\zeta^2}} \qquad \bar{g}_{i_+} = \bar{g}_i$$

$$\lambda_+ = \frac{\nu}{\mu} = \frac{p}{\mu^2} = \lambda \frac{h_{11}}{\mu^2}$$

$\lambda_{i+} = \lambda_i \frac{h_\perp}{\mu}$ $\quad x:\lambda \quad y:h_{\parallel}$

$a: \chi$



After s refinements / rounds,

$q = K^s$

$[A_k, B_k] = \left[-\alpha + \frac{2(k-1)\alpha}{q}, -\alpha + \frac{2k\alpha}{q}\right]$

midpoint of $\rightarrow$ is $\theta_k = \frac{2k-1-q}{q}\alpha$.

angular half-width $\omega = \frac{\alpha}{q} = \frac{\alpha}{K^s}$

$A_k = \theta_k - \omega \qquad B_k = \theta_k + \omega$

$\ell_k = \frac{\tan A_k}{\tan\alpha} \qquad r_k = \frac{\tan B_k}{\tan\alpha}$ $\qquad G(\alpha) := \frac{2\alpha\sec^2\alpha}{\tan\alpha} = \frac{2\alpha}{\sin 2\alpha}$.

active: $\lambda_+/\lambda \le \frac{d\delta^2}{4} \le \frac{d\,G(\alpha)^2}{K^{2s}}$ $\qquad \delta := r - \ell < \frac{2}{\sqrt{d}}$

$\delta = \frac{\tan B_k - \tan A_k}{\tan\alpha}$

inactive: $\lambda_{i+}/\lambda_i \le \frac{d}{d-1}$ $\qquad \le \frac{\sec^2\alpha\,(B-A)}{\tan\alpha} = \frac{\sec^2\alpha}{\tan\alpha}\frac{2\alpha}{K^s}$

$= \frac{2\,G(\alpha)}{K^s}$

Ratios $R_{act} = \sqrt{\frac{\lambda_+}{\lambda}}$ $R_{ina} = \sqrt{\frac{\lambda_{i+}}{\lambda_i}}$

$R_{act} \le \frac{\sqrt{d}\, G(\alpha)}{K^s}$

$R_{ina} \le \sqrt{\frac{\alpha}{\alpha-1}}$

A cycle of ds epochs.

$$\frac{\lambda_i^{next}}{\lambda_i} \le R_{act}^2 \cdot \left(R_{ina}^2\right)^{d-1} \qquad C_d := \sqrt{d}\left(\frac{d}{d-1}\right)^{\frac{d-1}{2}} \le \sqrt{ed}$$

$$= \left(\frac{C_d\, G(\bar\alpha)}{K^s}\right)^2 .$$

$$R_{i,ds} \le \frac{C_d\, G(\bar\alpha)}{K^s} \qquad \bar\alpha(\alpha) := \arctan\left[\left(\frac{d}{d-1}\right)^{\frac{d-1}{2}} \tan\alpha\right]$$

↑ worst case of α in a cycle.

$N = \frac{T}{ds}$. N cycles.

after N cycles: $\tan\alpha_N \approx \left(\frac{C_d}{K^s}\right)^N \tan\alpha_0$

$$\tan\alpha_T \approx \left(\frac{C_d}{K^s}\right)^{\frac{T}{ds}}$$

asymptotic penalty for compressing the exact info with an EC

$$\alpha_T = O\left(\exp\left[-\left(\frac{\log K}{d} - \frac{\log C_d}{ds}\right)T\right]\right)$$

$\le \frac{\log C_d}{d} \le \frac{\ln 2}{2}$

↑

at $d=2$.

↓

Never do MVEE, retain exact angular info.

$$\alpha_T \sim K^{-T/d} = \exp\left(-\frac{\log K}{d} T\right)$$

$$T(\varepsilon) \leq \frac{d_S}{s\log K - \log C_d} \log\frac{1}{\varepsilon} + O(1)$$

Initial cases.

$$\alpha = \frac{\pi}{2} - \varepsilon \qquad b_d = \left(\frac{d}{d-1}\right)^{\frac{d-1}{2}} \to \sqrt{e}.$$

$$\bar{\alpha} \approx \frac{\pi}{2} - \frac{\varepsilon}{b_d} < \frac{\pi}{2} - \frac{\varepsilon}{\sqrt{e}}$$

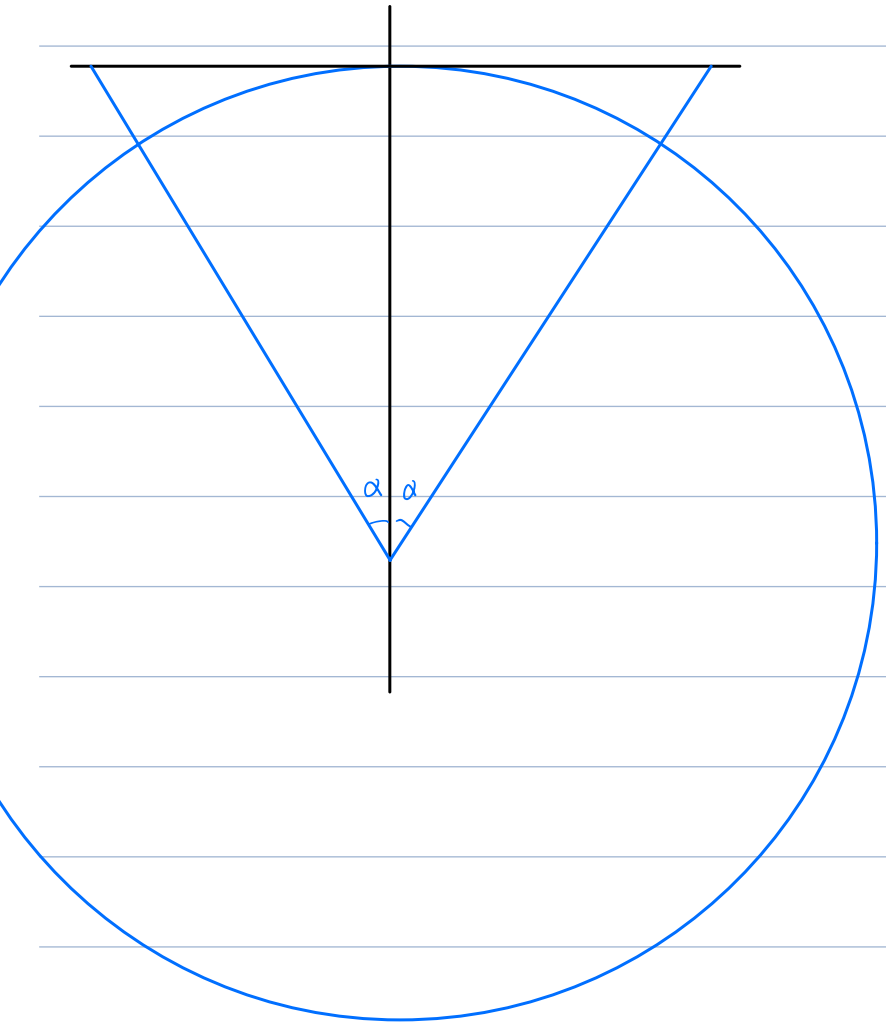
$$G(\bar{\alpha}) \approx \frac{\pi b_d}{2\varepsilon} < \frac{\pi\sqrt{e}}{2\varepsilon}.$$

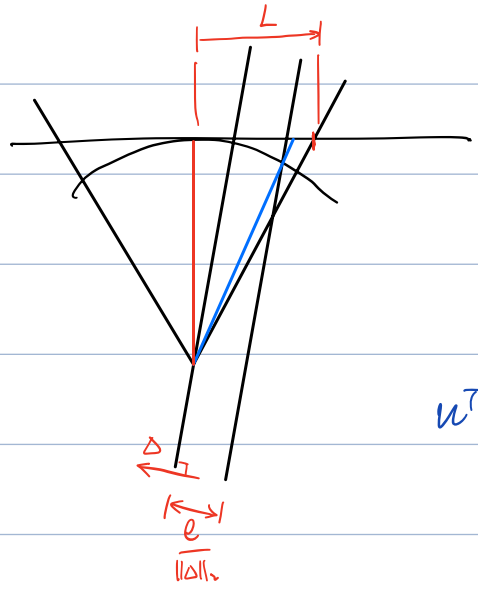
$$C_d = \sqrt{d}\, b_d < \sqrt{ed}.$$

$K^s > C_d\, G(\bar{\alpha})$

$$K^s \gtrsim \frac{\pi e\sqrt{d}}{2\varepsilon} = \frac{\pi e\sqrt{d}}{2(\frac{\pi}{2}-\alpha)}$$

$$s \gtrsim \left\lfloor \log_K \left[C_d\, G(\bar{\alpha}(\alpha)) \right] \right\rfloor + 1$$


α
α



$$\frac{(c+Fy)^T \Delta}{\sqrt{1+\|y\|^2}} \geq -e.$$

query plane.

$$\Downarrow$$

$$R = \|u_{\parallel}\| \geq \cos\bar{\alpha}$$

$$(c+Fy)^T \Delta \geq -e\sqrt{1+\|y\|^2} \geq -e\sqrt{1+L^2}$$

$$u^T(q_w - q_{w+1}) = -R\|q_w - q_{w+1}\| \sin(\phi - b) \geq -e\|q_w - q_{w+1}\|$$

$$u^T(q_w - q_\ell) \geq -e\|q_w - q_\ell\|, \quad e = \frac{\epsilon/\tau}{\sqrt{\lambda_{\min}(V_t)}}$$

$$\sin(\varpi) = \sin(\phi - b) \leq \frac{e}{R} \leq \frac{e}{\cos\bar{\alpha}} = \sec\bar{\alpha}\, e$$

α_* : worst angle

$$\frac{\sin(\varpi)}{\varpi} \geq \frac{\sin(2\bar{\alpha})}{2\bar{\alpha}} \quad L_*^{2\bar{\alpha}} : \ w \leq L_* e$$

$$\varpi \leq G(\bar{\alpha})\sin(\varpi) \leq G(\bar{\alpha})\sec\bar{\alpha}\, e.$$

a_* : max contraction multiplier in active direction

b_* :

$$\Lambda_i := \max_{k \neq i} \lambda_k$$

$$\sin(\phi - b) = \sin\phi\cos b - \cos\phi\sin b = \frac{x\cos b - \sin b}{\sqrt{1+x^2}} \quad ②$$

$$R_{i,\min}(u)\cdot\sin(\phi - b) = ① ② \leq h_t \cdot \|n_i(b)\|_{V_t^{-1}}$$

$$\sum_{k\neq i} \frac{y_k^2}{\lambda_k} \leq 1 - \frac{x^2}{\lambda_i}$$

$$\sum_{k\neq i} y_k^2 \leq \Lambda_i\left(1 - \frac{x^2}{\lambda_i}\right)$$

$$R_i(u) = \sqrt{\frac{1+x^2}{1+\|y\|^2}} = \sqrt{\frac{1+x^2}{1+x^2+\sum_{k\neq i} y_k^2}}$$

$$\geq R_{i,\min}(u) = \sqrt{\frac{1+x^2}{1+x^2\Lambda_i(1-\lambda_i^{-1})}} \quad ①$$

$$u^T n_i(b) \leq h_t \cdot \|n_i(b)\|_{V_t^{-1}}$$

$$R_i(u)\cdot\sin(\phi - b) = u_{proj}^T n_i(b) = u^T n_i(b) \leq h_t \cdot \|n_i(b)\|_{V_t^{-1}}$$

$$\phi - b \leq \arcsin\left(\frac{h_t \cdot L_{t,i}(b)}{R_i(u)}\right)$$

$\|u_{proj}\| = R_i(u)$

$\phi - b$

$\|n_i(b)\|_{V_t^{-1}}$

$n_i(b)$ (unit)

$-\sin b\, c + \cos b\, g_i$